Band-1 QUE: Arithmetic Surface / Hecke-Mass Remainder Probe (Collapse 2.36)

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Band-1 QUE — One Arithmetic Surface + Hecke-Mass Remainder (Lit + Statement Lock)

Honesty stamp. This note does **not** claim Quantum Unique Ergodicity in full generality. It does **not** claim holomorphic QUE on compact arithmetic surfaces. It does **not** claim a Clay-millennial solution or a Theory of Everything. **GATE-04** is locked out of scope. Lit lock only — **no** Maass-form / spectral numerics on box.

> **Locked PROVED (Band-1 surface):** Hecke–Maass QUE on $X = \mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$ — **Lindenstrauss** (quantum limits $= c \cdot \mu_{\mathrm{hyp}}$) + **Soundararajan 2010** ($c = 1$; **no escape of mass** / Hecke-mass remainder closed).
> **Locked PROVED (same surface, holomorphic):** mass equidistribution for holomorphic Hecke eigencuspforms — **Holowinsky–Soundararajan 2010**.
> **Locked OPEN:** holomorphic QUE on **compact** arithmetic hyperbolic surfaces — **OPEN**.

Abstract

Rudnick–Sarnak conjectured Quantum Unique Ergodicity (QUE): on hyperbolic surfaces of negative curvature, every sequence of L^2-normalized Laplacian eigenfunctions equidistributes to hyperbolic area in the large-eigenvalue limit (stronger than density-one Quantum Ergodicity). In the **arithmetic** setting one restricts to **Hecke** eigenforms. **Lindenstrauss** proved arithmetic QUE for Hecke–Maass forms on **compact** arithmetic surfaces, and classified quantum limits on the noncompact modular surface up to a possible escape-of-mass constant $c \in [0, 1]$. **Soundararajan** eliminated escape of mass on $\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$, completing Hecke–Maass QUE there. **Holowinsky–Soundararajan** proved the holomorphic mass-equidistribution conjecture on the same noncompact surface. The classical **remaining** arithmetic-surface case is **holomorphic QUE on compact** arithmetic surfaces (Nelson: reduces to quadratic Hecke-sum bounds).

This Band-1 probe locks **one** arithmetic surface with its Hecke-mass remainder closed: the modular surface. It does **not** elevate that to full QUE.

Problem (collapse Band-2 item 2.36 → Band-1 lit+statement)

1. Ledger QUE / arithmetic Hecke QUE; Lindenstrauss / Soundararajan / Holowinsky–Soundararajan orbit.
2. Lock **one** literature-proved arithmetic surface as the Band-1 claim surface (with Hecke-mass / escape remainder closed).
3. Name **one remaining open class** (compact holomorphic).
4. Keep Clay / ToE / GATE-04 / "full QUE" off the claim surface.
5. Lit lock enough — no box spectral numerics.

Canonical statement

Let $X = \Gamma\backslash\mathbb{H}$ be a finite-area hyperbolic surface and $\{\phi_j\}$ an L^2-normalized orthonormal basis of Maass cusp forms (Laplacian eigenfunctions). Write

$$\mu_j \;=\; |\phi_j(z)|^2 \, \frac{dx\,dy}{y^2}.$$

QUE (Rudnick–Sarnak): every weak-$*$ limit of μ_j as $\lambda_j \to \infty$ equals the normalized hyperbolic area μ_{hyp}. **Arithmetic QUE** asks the same for eigenfunctions that are also eigenfunctions of all Hecke operators. The **holomorphic** analogue replaces Maass forms by holomorphic Hecke eigencuspforms of weight $k \to \infty$, with mass measure proportional to $|f(z)|^2 y^k \, \frac{dx\,dy}{y^2}$.

Hecke-mass / escape-of-mass remainder (noncompact): on $X = \mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$, Lindenstrauss showed any quantum limit is $c \cdot \mu_{\text{hyp}}$ with $0 \le c \le 1$; the remainder is to prove $c = 1$ (no mass escapes into the cusp).

Literature ledger (web check; not re-proved on box)

Year	Result	Authors	Status
1994	QUE conjecture formulated (arithmetic motivation)	Rudnick–Sarnak	conjecture
1970s–80s	Quantum Ergodicity (density one)	Shnirelman; Colin de Verdi‘ere; Zelditch	**proved** (QE)
1995	Density-one mass equidistribution for holomorphic Hecke forms	Luo–Sarnak	**proved** (QE-type)
2002	Watson triple-product formula (QUE $\Leftarrow$ subconvexity)	Watson	tool
2006	Arithmetic QUE for Hecke–Maass on **compact** arithmetic surfaces; on modular surface limits $= c \cdot \mu$	**Lindenstrauss**, *Ann. of Math.*	**proved**
2010	**No escape of mass** on $\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H} \Rightarrow$ Hecke–Maass QUE	**Soundararajan**, *Ann. of Math.* 172	**proved** — **Band-1 mass remainder**
2010	Holomorphic mass equidistribution on modular surface	**Holowinsky–Soundararajan**, *Ann. of Math.* 172	**proved** (noncompact holomorphic)
2011	Survey: recent progress on QUE	Sarnak, *Bull. AMS*	survey
2020/22	Compact holomorphic QUE $\Leftarrow$ nontrivial quadratic Hecke sums; twisted base-change variant	Nelson, arXiv:2001.08704 / *IMRN*	criterion + partial; full compact holomorphic still **OPEN**
—	Holomorphic QUE on **compact** arithmetic surfaces	—	**OPEN**

What is proved vs open (statement lock)

Literature-proved (cite carefully)

1. **QE (not unique)** — Shnirelman / Colin de Verdi'ere / Zelditch: density-one equidistribution.
2. **Compact arithmetic Hecke–Maass QUE** — Lindenstrauss (2006): measure rigidity for Hecke–invariant quantum limits on compact arithmetic hyperbolic surfaces.
3. **Band-1 surface + Hecke-mass remainder:** $X = \mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$. Lindenstrauss: quantum limits of Hecke–Maass forms are $c \cdot \mu_{\mathrm{hyp}}$, $0 \le c \le 1$. **Soundararajan (2010):** $c = 1$ (no escape of mass), completing **Hecke–Maass QUE** on the modular surface. **This is the collapse-2.36 "new arithmetic surface / Hecke-mass remainder" locked here.**
4. **Holomorphic on the same noncompact surface** — Holowinsky–Soundararajan (2010): synthesizing complementary weak-subconvexity and sieve methods; relies on Deligne's Ramanujan bound for holomorphic forms.
5. **Dihedral / CM specials** — Sarnak; **density-one holomorphic** — Luo–Sarnak.
6. **Higher-rank arithmetic advances** — Silberman–Venkatesh et al. (restricted locally symmetric settings; not full general QUE).

Locked OPEN (named remaining class)

1. **Holomorphic QUE on compact arithmetic hyperbolic surfaces** — **OPEN**. Holowinsky's shifted-convolution / sieve method uses Fourier expansions unavailable in the nonsplit compact setting; Soundararajan's weak subconvexity alone does not close uniqueness. Nelson reduces the problem to nontrivial bounds on Hecke eigenvalues summed over quadratic progressions (Luo–Sarnak-type criterion without Fourier expansions) and proves a twisted base-change variant — **not** the full compact holomorphic theorem.
2. **Effective / power-saving Hecke-mass remainders** in full strength — **OPEN / CONDITIONAL** (qualitative QUE $\neq$ uniform power-saving rates; strong subconvexity typically required).
3. **Full non-arithmetic QUE** on general negatively curved surfaces — **OPEN** (Anantharaman: positive entropy of quantum limits; not uniqueness).

Explicit non-claims

- Full QUE **not** claimed (all negatively curved surfaces / all eigenstates).
- Compact holomorphic arithmetic QUE **not** claimed.
- Power-saving effective rates **not** claimed in full generality.
- No Clay; no ToE; **GATE-04 LOCKED**.
- No box Maass / holomorphic-form numerics presented as evidence.

Why Band-1 vs full QUE

- **Band-1 (this probe):** honest lit+statement lock for **one** arithmetic surface ($X = \mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$) with the Hecke-mass / escape-of-mass remainder **closed**, plus holomorphic QUE on that same noncompact surface as classical companion.
- **Band 2–3:** compact holomorphic arithmetic QUE; effective rates.
- **Band 4 / open-ended:** unrestricted QUE on general negatively curved manifolds.
- **Collapse note:** "Lindenstrauss / Soundararajan orbit" — reflected honestly as the modular-surface completion; compact holomorphic remainder open.

Box note

No Maass-form computation, no Hecke-eigenvalue search, no spectral-side numerics on box. Dash rule: **lit lock is enough**.

Deliverables

- Lab: `experiments/band1/que_arithmetic_surface/` (`literature_ledger.json`, `results.json`, `SUMMARY.txt`, `METHOD.md`, `README.md`, local `next_wave.md`).
- Paper/PDF: `papers/2026-09-11_Band1_QUE_arithmetic_surface_probe.{md,pdf}` → `DOWNLOAD_PDFs` + Mac `Downloads/DOWNLOAD_PDFs_2026-09-11/`.
- House `next_wave.md` updated.

References (primary)

1. Z. Rudnick, P. Sarnak, *The behaviour of eigenstates of arithmetic hyperbolic manifolds*, Comm. Math. Phys. **161** (1994).
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7. T.C. Watson, *Rankin triple products and quantum chaos*, Ph.D. thesis, Princeton (2002).
8. P.D. Nelson, *Quadratic Hecke sums and mass equidistribution*, Int. Math. Res. Not. (2022); arXiv:2001.08704.